GradientCIO

PRIVATE MARKETS PORTFOLIO CONSTRUCTION

# Northstar Private Markets Program

Portfolio construction review · prepared for the Investment Committee

AS OF
2026-06-30

PRIVATE-MARKETS NAV
$230.0M

UNFUNDED COMMITMENTS
$92.0M

REVIEW
Program construction

# 01 Executive Decision

Recommendation and committee action

**Pacing and concentration need monitoring**

RECOMMENDATION

We recommend a two-year $80.0M commitment program that shifts new commitments toward infrastructure and secondary private equity while keeping private-markets exposure inside the 15.0%–30.0% policy range S3 S6. Committee action requested: approve the program ceiling and first-year pacing, subject to manager diligence and quarterly liquidity tests S3 S7.

| PRIVATE-MARKETS WEIGHT | UNFUNDED COMMITMENTS | LIQUID COVERAGE | POLICY STATUS |
|---|---|---|---|
| 23.0% | $92.0M | 4.8x | Near limit |
| Policy range 15.0%–30.0% S3 | 9.2% of NAV S4 | Governed base case S3 | Manager concentration S3 |

**Recommendation and action requested**

We recommend a two-year $80.0M commitment program. Committee action requested: approve a $45.0M first-year ceiling and require completed diligence before each commitment S3 S6 S7.

DECISION SUMMARY

| DECISION ELEMENT | PROPOSED ACTION | EVIDENCE | STATUS |
|---|---|---|---|
| Program size | $80.0M over two years | S6 | Proposed |
| First-year ceiling | $45.0M | S6 | Proposed |
| Policy fit | Remain within 15.0%–30.0% | S3 | Compliant |

**Conditions and limits**

Each manager requires completed operational and GIPS diligence; quarterly pacing proceeds only while governed liquidity coverage remains at or above the returned policy threshold S3 S7.

**Key judgment — Pacing**

Observation: Commitment pacing and manager concentration both need monitoring S1. Why it matters: The $45.0M first-year ceiling keeps the program inside its liquidity limits. Uncertainty: Capital calls depend on manager deployment speed. What would change the view: Calls running ahead of the pacing plan.

**Key judgment — Sleeve size**

Observation: Private markets are 23.0% of total NAV S2. Why it matters: New commitments are sized against this weight and the unfunded balance. Uncertainty: Private valuations lag public markets. What would change the view: A valuation update that moves the weight by more than 2pp.

**Key judgment — Policy limit**

Observation: The private-markets allocation is near its policy limit S3. Why it matters: Each new commitment must be checked against the limit before signing. Uncertainty: The limit test uses lagged valuations. What would change the view: Distributions that bring the weight back toward target.

## 02 Mandate and Construction Objective

Policy and program constraints

| | |
|---|---|
| Portfolio | **Northstar Total Fund** S2 |
| Objective | **Diversified private-markets program** S7 |
| Horizon | **10 years** S5 |
| Target range | **15.0%–30.0% of total NAV** S3 |
| Base currency | **USD** S2 |
| Data scope | **Illustrative sample data** S1 S1 |

GOVERNED FRAMEWORK

| AREA | OBJECTIVE OR LIMIT | OBSERVED | STATUS | SOURCE |
|---|---|---|---|---|
| Private markets | 15.0%–30.0% | 23.0% | Compliant | S3 |
| Single manager | At or below 12.0% | 11.4% | Near limit | S3 S4 |
| Unfunded ratio | At or below 12.0% | 9.2% | Compliant | S3 |

## 03 Current Private Markets Portfolio

NAV, commitments and valuation dates

PRIVATE-MARKETS NAV BY STRATEGY



### CURRENT PROGRAM SNAPSHOT

| STRATEGY | NAV | UNFUNDED | COMMITMENT | % NAV | VALUATION DATE |
|---|---|---|---|---|---|
| Private equity | $120.0M | $42.0M | $210.0M | 12.0% | 2026-03-31 S4 |
| Private credit | $70.0M | $30.0M | $125.0M | 7.0% | 2026-03-31 S4 |
| Infrastructure | $40.0M | $20.0M | $75.0M | 4.0% | 2026-03-31 S4 |

### ILPA COMPANY COVERAGE

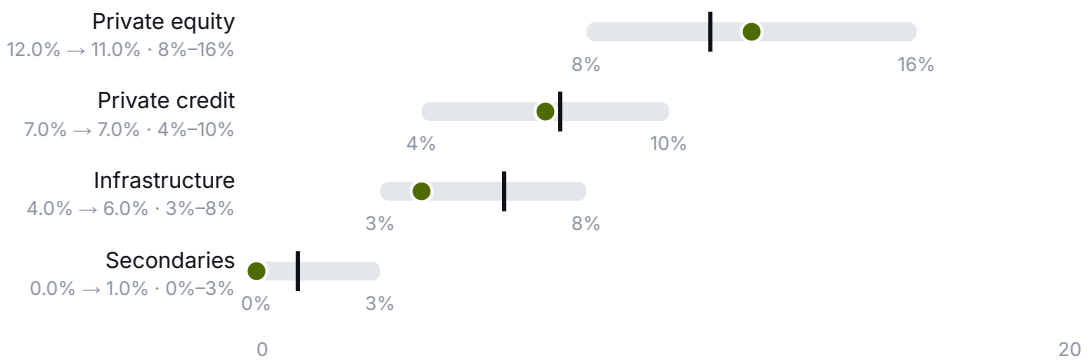
| DATA STATUS | MAPPED COMPANIES | VALUED COMPANIES | ASSESSED / TOTAL | AGGREGATE COMPLETENESS | SELECTED ARTIFACTS | REPORTING DATE | DRAFT LABEL |
|---|---|---|---|---|---|---|---|
| Incomplete | 2 | 2 | 2 / 3 | Incomplete — one unknown-grain holding excluded | 1 | 2026-06-30 | Draft — July 22, 2026 S9 |

All reported private-market NAVs are dated 2026-03-31 and therefore lag the report date; unfunded amounts are shown separately and are not included in NAV. Company look-through is a non-additive risk overlay; the partial server aggregates assess two of three holdings, with one unknown grain excluded S4 S9.

## 04 Target Portfolio Structure

The proposed mix reduces concentrated risk while preserving mandate exposure

### CURRENT WEIGHT VERSUS POLICY RANGE




### CURRENT AND TARGET MIX

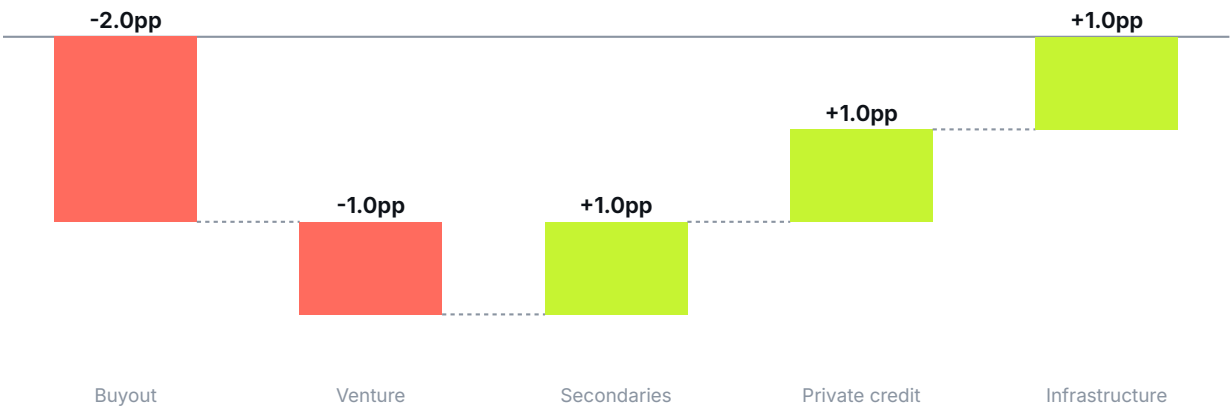



### VINTAGE-YEAR MIX




Governed vintage evidence and proposed commitment pacing S6 S7.

### TARGET ALLOCATION CHANGES (PERCENTAGE POINTS)




### CURRENT VERSUS TARGET

| SEGMENT | CURRENT | TARGET | CHANGE | RANGE / CONSTRAINT | RATIONALE |
|---|---|---|---|---|---|
| Private equity | 12.0% | 11.0% | -1.0pp | 8.0%–16.0% | Use slower new pacing S3 S6 |
| Private credit | 7.0% | 7.0% | 0.0pp | 4.0%–10.0% | Maintain income role S5 S7 |
| Infrastructure | 4.0% | 6.0% | +2.0pp | 3.0%–8.0% | Direct new commitments here S5 S6 |
| Secondaries | 0.0% | 1.0% | +1.0pp | At or below 3.0% | Add vintage diversification S6 S7 |

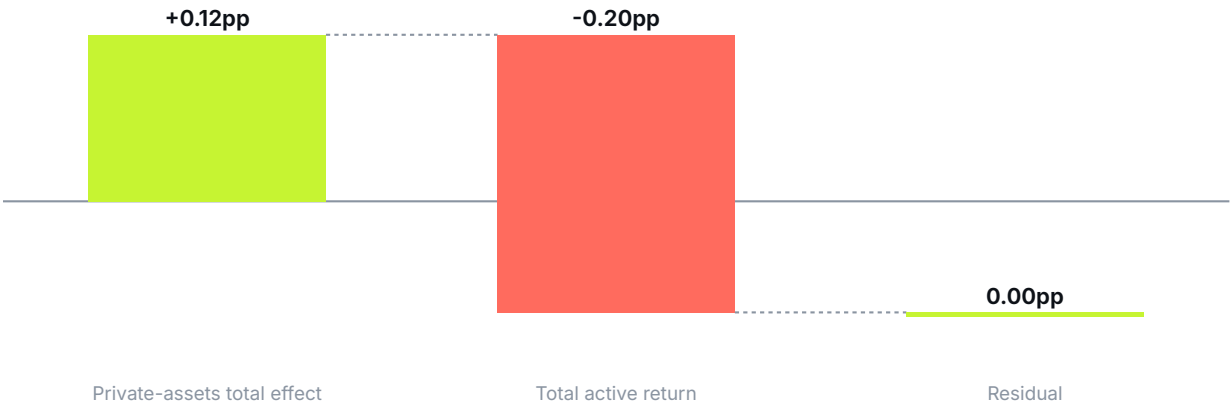
> **Analysis — Why this target**
>
> Observation: New pacing increases two roles, infrastructure and secondaries, versus the current mix S6. Why it matters: The broader mix remains inside policy boundaries S3. Uncertainty: CMA, regime and manager-finding coverage remain incomplete S7. What would change the view: Liquidity coverage below policy or completed records rejecting either role S3 S7.

Targets are a committee proposal from the illustrative mandate and returned policy headroom; they are not changes saved to GradientCIO S3 S7.

# 05 Historical Performance and Attribution

Returned portfolio evidence; sleeve gaps remain explicit

### GOVERNED ATTRIBUTION EFFECTS




Returned attribution effects; residual is shown explicitly S5.

### STANDARD PERIODS

| PERIOD | PORTFOLIO / SLEEVE | BENCHMARK | EXCESS | COVERAGE |
|---|---|---|---|---|
| 1Y total portfolio | 7.2% | 7.6% | -0.4pp | Complete S8 |
| 3Y total portfolio (ann.) | 6.8% | 6.5% | +0.3pp | Complete S8 |

### ATTRIBUTION EFFECTS

| EFFECT | CONTRIBUTION | SOURCE |
|---|---|---|
| Private-assets total effect | +0.12pp | S8 |
| Total active return | -0.20pp | S8 |
| Residual | 0.00pp | S8 |

Governed attribution method and reconciliation metadata follow S5.

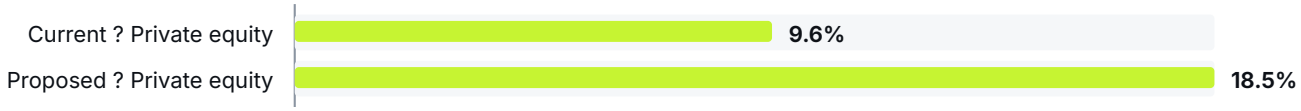
### METHOD

| | |
|---|---|
| Method | **Brinson-Fachler** S8 |
| Linking | **Symmetric Carino** S8 |
| Basis | **Net, monthly linked** S8 |
| Tolerance | **1 bp; reconciled** S8 |

# 06 Forward Return and Risk

Governed assumptions frame tradeoffs rather than forecasts

### FORWARD EXPECTED-RETURN COMPARISON




Governed forward assumptions; assumptions, not forecasts S6.

| | |
|---|---|
| Assumption set | **2026 Strategic CMA** S5 |
| Regime | **Baseline** S5 |
| Horizon | **10 years** S5 |
| Currency | **USD** S5 |
| Context fingerprint | **test-pm-forward-2026** S5 |

### PRIVATE-MARKET FORWARD ASSUMPTIONS

| SEGMENT | EXPECTED RETURN | EXPECTED VOLATILITY |
|---|---|---|
| Private equity | 9.6% | 18.5% |
| Private credit | 7.1% | 9.2% |
| Infrastructure | 7.8% | 11.8% |

**Analysis — Assumptions frame tradeoffs**

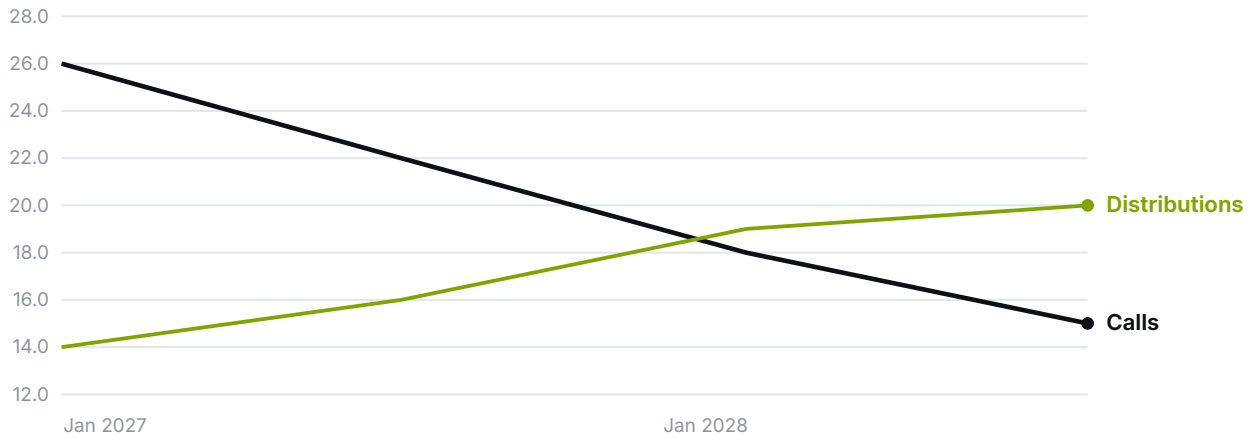
Observation: Private equity has higher expected return and volatility than every other selected segment S5. Why it matters: Segment sizing balances return potential against modeled risk. Uncertainty: These long-horizon assumptions are not manager forecasts. What would change the view: A governed assumptions refresh reversing either relative ranking S5.

The figures are long-horizon assumptions under the selected context, not forecasts or guarantees S5.

# 07 Commitments, Pacing and Cash Flow

Returned portfolio commitment schedule

### COMMITMENT PACING (USD MILLIONS)




Illustrative projection for committee review S6.

### PROJECTED CALLS AND DISTRIBUTIONS

| PERIOD | CALLS | DISTRIBUTIONS |
|---|---|---|
| 2026 H2 | USD 26,000,000 | USD 14,000,000 |
| 2027 H1 | USD 22,000,000 | USD 16,000,000 |
| 2027 H2 | USD 18,000,000 | USD 19,000,000 |
| 2028 H1 | USD 15,000,000 | USD 20,000,000 |

### NEAR-TERM PACING

| PERIOD | CALLS | DISTRIBUTIONS | NET CASH FLOW | BASIS | SOURCE |
|---|---|---|---|---|---|
| 2026 H2 | $26.0M | $14.0M | -$12.0M | Returned projection | S6 |
| 2027 H1 | $22.0M | $16.0M | -$6.0M | Returned projection | S6 |
| 2027 H2 | $18.0M | $19.0M | +$1.0M | Returned projection | S6 |

# 08 Liquidity and Denominator Risk

Coverage remains adequate under the sample stress case

### GOVERNED SCENARIO COVERAGE




Denominator stress is unavailable S3 S6.

### LIQUIDITY COMPOSITION




### LIQUIDITY PROFILE

| BUCKET | % NAV | AMOUNT | REQUIREMENT | STATUS |
|---|---|---|---|---|
| Daily to quarterly | 44.0% | $440.0M | At least 25.0% | Compliant |
| Over two years / locked | 23.0% | $230.0M | At or below 30.0% | Compliant |

| | |
|---|---|
| Liquid assets | **$440.0M** S3 |
| Unfunded commitments | **$92.0M** S4 |
| Base coverage | **4.8x** S3 |
| Near-term projected calls | **$48.0M** S6 |

**Analysis — Coverage supports pacing**

Observation: Base liquid coverage is 4.8x against unfunded commitments S3 S4. Why it matters: Current resources support the staged pacing case. Uncertainty: A governed denominator-stress result remains unavailable for this portfolio. What would change the view: Lower resources, higher calls, or stress results below the policy threshold S3 S6.

### SUPPORTED SCENARIOS

| SCENARIO | LIQUID RESOURCES | FUNDING NEED | COVERAGE | STATUS |
|---|---|---|---|---|
| Governed base case | $440.0M | $92.0M | 4.8x | Compliant |
| Denominator stress | Not available — no governed stress result | Not available | Not available | Not assessed |

## 09 Diversification and Manager Diligence

Concentration and pre-commitment conditions

### DIVERSIFICATION

| AXIS | LARGEST EXPOSURE | SHARE | LIMIT | STATUS | SERVER METRIC / BASIS |
|---|---|---|---|---|---|
| Manager | Example Private Manager A | 11.4% | 12.0% | Near limit | Commitment exposure; additive S4 |
| Strategy | Private equity | 52.2% of private NAV | Not assessed | Not assessed | Commitment exposure; additive S4 |
| Portfolio company | illustrative Company A | 24.0% of valued company NAV | Not assessed | Partly available | Server company_concentration; canonical NAV; non-additive S9 |
| Sector | Technology | 41.0% of valued company NAV | Not assessed | Partly available | Server sector_concentration; canonical NAV; non-additive S9 |

### PORTFOLIO-COMPANY EVIDENCE

| COMPANY | CANONICAL NAV | EXACT ILPA VALUE | VALUATION / REPORTING DATE | PROJECTED RANGE | LEVERAGE / COVERAGE | COVENANT STATUS | OPERATING-QUALITY EVIDENCE | RECONCILIATION | SOURCE |
|---|---|---|---|---|---|---|---|---|---|
| illustrative Company A | $24.0M | FMV $24.0M in USD | 2026-06-30 / 2026-06-30 | Leverage 3.0–4.0x | Interest coverage 2.0–3.0x | In compliance | Revenue growth 5%–10% | Matched | S9 |
| illustrative Company B | $18.0M | FMV $18.0M in USD | 2026-06-30 / 2026-06-30 | Leverage 4.0–5.0x | Interest coverage 1.5–2.0x | Watch | EBITDA margin 15%–20% | Matched | S9 |

Selected artifact illustrative-ilpa-artifact-test, version 1; parser ilpa_portfolio_company_draft_2026_07_22 version 1.0.0; projection registry private-markets-holding-exact-to-range-v1, checksum aaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa. Draft — July 22, 2026. Canonical values, exact ILPA source values and projected Gradient ranges remain separate S9.

### MANAGER EVIDENCE

| MANAGER / FUND | EXPOSURE | DILIGENCE STATUS | OPEN FINDINGS | NEXT STEP |
|---|---|---|---|---|
| Example Private Manager A | $114.0M | Current S7 | 1 Medium S7 | Close valuation-control follow-up |
| Proposed infrastructure manager | Not funded | Not assessed S7 | Not assessed | Run manager comparison and ODD |

## 10 Proposed Program and Implementation

Phased commitments with decision gates

### COMMITMENT PROGRAM

| PERIOD | SEGMENT | PROPOSED AMOUNT | FUNDING SOURCE | CONDITION | SOURCE |
|---|---|---|---|---|---|
| 2026 H2 | Infrastructure | $25.0M | Distributions and liquid reserve | Complete ODD | S6 S7 |
| 2027 H1 | Private credit | $20.0M | Distributions | Coverage remains compliant | S3 S6 |
| 2027 H2 | Secondaries | $20.0M | Distributions and cash | Committee checkpoint | S3 S6 |
| 2028 H1 | Reserve | $15.0M | To be determined | Evidence refresh | S6 |

### ALTERNATIVES CONSIDERED

| ALTERNATIVE | BENEFIT | COST / RISK | REASON NOT SELECTED |
|---|---|---|---|
| No new commitments | Maximum liquidity | Target exposure declines | Does not address program objective S3 S6 |
| Front-load commitments | Faster target progress | Higher near-term calls | Coverage sensitivity S3 S6 |

- Phase each commitment after diligence clearance S7.
- Refresh pacing and policy liquidity quarterly S3 S6.
- Return to committee if manager concentration reaches its governed limit S3.

# 11 Risks, Open Items and Approvals

Conditions that govern implementation

### DECISION RISKS

| PRIORITY | RISK | EVIDENCE | MITIGATION / CONDITION | OWNER |
|---|---|---|---|---|
| High | Valuation and denominator lag | Q1 valuations S4 | Quarterly headroom test | Investment staff |
| Medium | Manager concentration | 11.4% vs 12.0% S3 S4 | Cap follow-on commitments | CIO |
| Medium | Near-term net calls | $18.0M through 2027 H1 S6 | Maintain liquidity gate | Treasury |

### OPEN ITEMS AND APPROVALS

| TIMING | ITEM | REQUIRED EVIDENCE | OWNER | STATUS |
|---|---|---|---|---|
| Before approval | Confirm first manager | Comparison and ODD | Private markets team | Open S7 |
| Within 30 days | Document pacing limits | Final program schedule | CIO | Open S6 |
| Next review | Refresh valuations | 2026 Q2 NAVs | Operations | Open S4 |

# 12 Coverage

| | | |
|---|---|---|
| **Portfolio and allocation** | **Available** | Illustrative sample data S2 |
| **Policy** | **Available** | Allocation, liquidity and concentration S3 |
| **Exposure** | **Available** | NAV, commitments and unfunded S4 |
| **ILPA portfolio-company look-through** | **Incomplete** | 2 of 3 holdings assessed; one unknown grain; Draft — July 22, 2026 S9 |
| **Forward assumptions** | **Available** | 2026 Strategic CMA S5 |
| **Commitments charts** | **Available** | Returned pacing projection S6 |
| **Manager diligence** | **Incomplete** | Proposed manager not assessed S7 |
| **Returns and attribution** | **Incomplete** | No sleeve-specific return S8 |
| **CMA consensus** | **Not available** | Optional context not returned S5 |
| **Macro regime** | **Not available** | Optional context not returned S5 |
| **Strategy Lab** | **Not available** | No selected series S8 |
| **Peer allocation context** | **Not available** | Optional context not returned S7 |
| **Transaction costs** | **Not available** | Not returned S8 |

# A Appendix — Sources and methods

Based on 9 of 10 evidence areas.

| TAG | EVIDENCE | SOURCE | AS OF | CHECK |
|---|---|---|---|---|
| S1 | Org capabilities | list_organizations; get_gradient_capabilities | 2026-06-30 | Passed |
| S2 | Portfolio / allocation | list_portfolios; get_portfolio_structure | 2026-06-30 | Passed |
| S3 | Policy assessment | check_portfolio_policy | 2026-06-30 | Passed |
| S4 | Private-market exposure | get_portfolio_exposure | 2026-03-31 | Passed |
| S5 | Forward assumptions | get_capital_market_assumptions | 2026-06-15 | Passed |
| S6 | Commitments and pacing | get_portfolio_exposure | 2026-06-30 | Passed |
| S7 | Manager / mandate | get_gradient_capabilities | 2026-06-30 | Advisory |
| S8 | Returns / attribution | get_portfolio_historical_returns; get_portfolio_attribution | 2026-06-30 | Incomplete |
| S9 | ILPA company look-through | get_private_markets_holdings | 2026-06-30 | Partly available |

Illustrative, Gradient Maintained — demo data, not the client's holdings or managers. All organizations, portfolios, managers and results in this public example are illustrative.

### SOURCE AUDIT TRAIL

| TAG | KEY PARAMETERS | DIGEST |
|---|---|---|
| S1 | org=test | sha256:aaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa |
| S2 | allocation_tree | sha256:aaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa |
| S3 | portfolio=test | sha256:aaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa |
| S4 | all pages | sha256:aaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa |
| S5 | baseline | sha256:aaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa |
| S6 | commitments | sha256:aaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa |
| S7 | illustrative | sha256:aaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa |
| S8 | all sections | sha256:aaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa |
| S9 | paged; server aggregates | sha256:aaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa |

Data scope for every source: Illustrative sample data. Each digest is the SHA-256 fingerprint of the exact data the report used.

# B Appendix — Server Metric Methods and Disclosures

### SERVER METRIC METHODS

| SOURCE | QUANTITY | METHOD / FORMULA | VERSION | BASIS / UNITS / TOLERANCE |
|---|---|---|---|---|
| S3 | Policy status | Governed portfolio assessment | portfolio-policy-v1 | Returned status |
| S4 | Exposure | Server exposure aggregation | portfolio-exposure-v1 | USD; unfunded separate |
| S5 | Forward statistics | Expected-statistics charts | chart-contract-v1 | 10-year assumptions |
| S9 | Company concentration | Server aggregates only | private-markets-holding-exact-to-range-v1 | Canonical NAV; non-additive; partial coverage |

Private-market valuations may lag the report date S4. Company look-through is non-additive to commitment exposure, and exact ILPA source money is not summed across currencies S9. The selected portfolio-company artifact is Draft — July 22, 2026, not a final ILPA release S9. Forward values are assumptions, not forecasts or guarantees S5. Portfolio Analytics and Strategy Lab are separate analytical scopes. This AI-assisted construction analysis is not investment, legal or compliance advice.